

Reference excerpt 02-021

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$$t_{ax} + T_c v_i = \max(CSA(t)) : t_{xi} + t_{xv} - T_c < t < t_{xi} + t_{cx} + T_c v_i = \min(CSA(t)) : t_{vi} + t_{vy} - T_c < t < t_{vi} + t_{vy} + T_c \quad (1)$$

Figure 1: The figure represents an illustrative sequence of three sonograms from a video-clip obtained by a transverse ultrasound scan of the IJV. In each picture it is visible the IJV (ellipses), the common carotid artery (circular) and the ECG trace (blue line). The active phase of the ECG is indicated by the red cursor. Below, it shows the IJV CSA trace obtained by measuring the CSA in cm^2 on each sonogram of the video-clip. The parameter ranges between 0 and 1 (typically 0.05) and serves to take account of heart rate variation (HRV) during acquisition. The t_{ai-1} is the instant corresponding to the wave a during the $(i-1)$ -th cycle, the parameters t_{xi} and t_{vi} are defined similarly. The parameters t_{ac} , t_{ax} , t_{xv} and t_{vy} represent the time interval between the waves a and c, c and x, x and v, v and y, respectively (see Fig. 2) and are measured by an operator over a selected cardiac cycle. The detection algorithm takes into account their possible modification during subsequent cycles, by means of the parameter (see Eq. (1)). Once identified the waves a_i , c_i , x_i , v_i and y_i on each cardiac cycle, the intervals t_{xi} and t_{vi} are calculated for each cardiac cycle. The parameters defined in